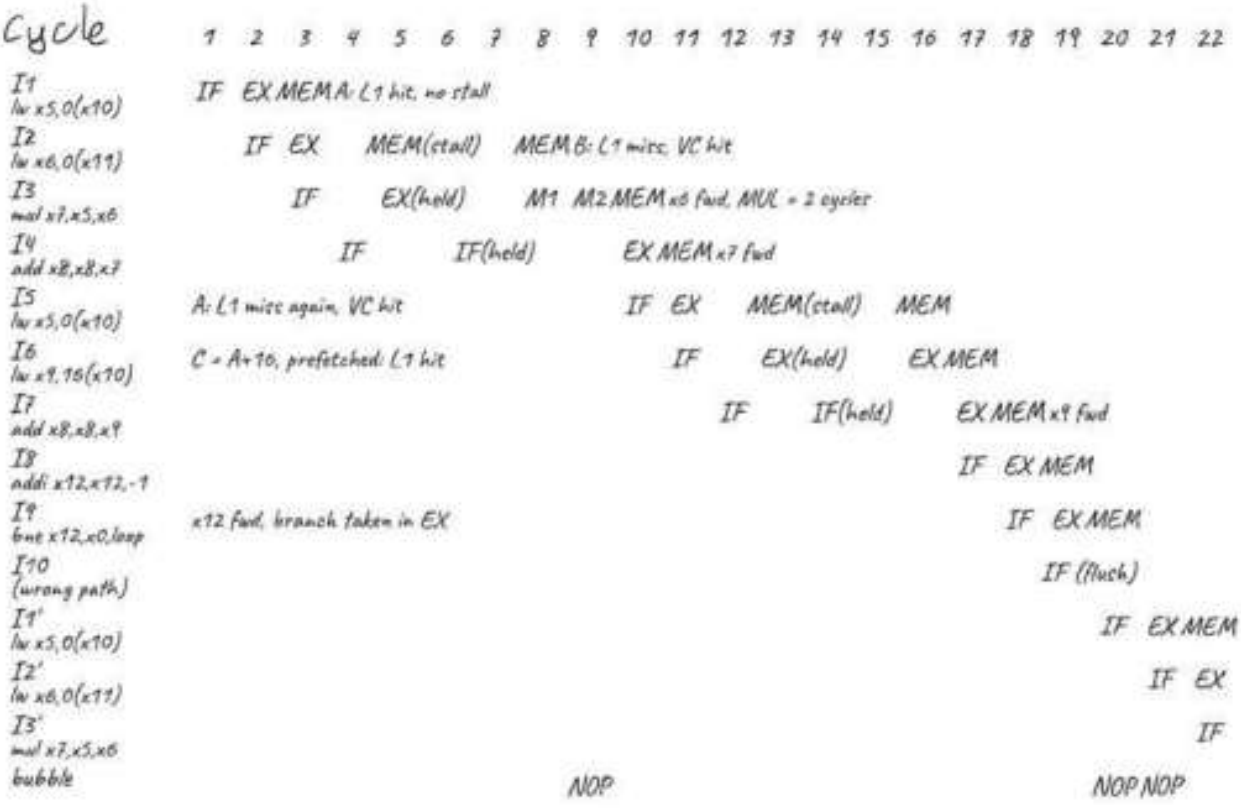
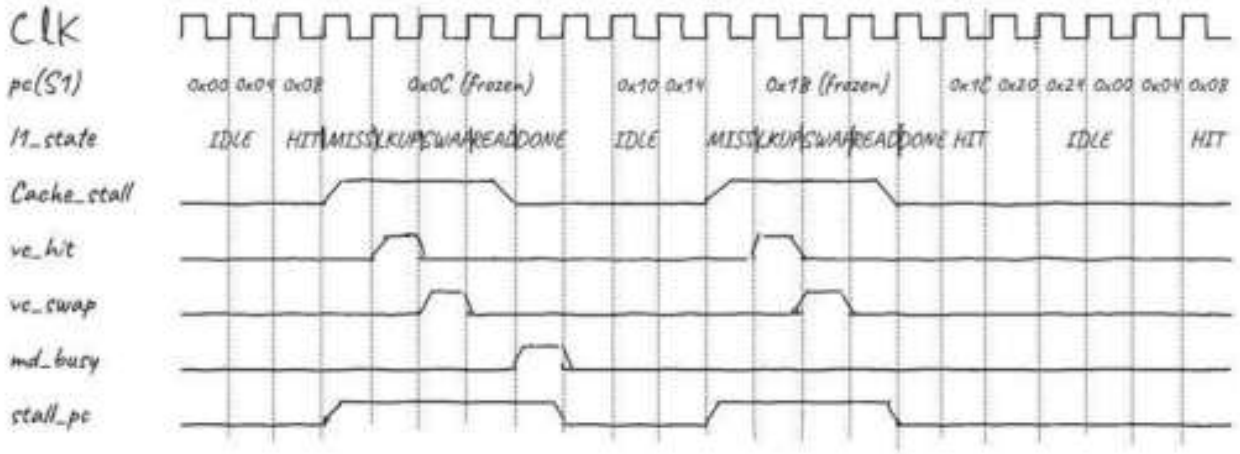
2. Timing Diagram (FULL PAGE)

Requirements:

- ≥ 20 cycles
- Must correspond to a **real instruction sequence**
- Must clearly show:
  - pipeline stages per cycle
  - MUL/DIV execution (multi-cycle if applicable)
  - stalls / forwarding / hazards

Idealized diagrams without hazards → not allowed


Cycle
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22
I1 lw x5,0(x10)
IF EX MEM A: L1 hit, no stall
I2 lw x6,0(x11)
IF EX MEM(stall) MEM B: L1 miss, VC hit
I3 mul x7,x5,x6
IF EX(hold) M1 M2 MEM x6 fwd, MUL = 2 cycles
I4 add x8,x8,x7
IF IF(hold) EX MEM x7 fwd
I5 lw x5,0(x10)
A: L1 miss again, VC hit
IF EX MEM(stall) MEM
I6 lw x9,16(x10)
C = A+16, prefetched: L1 hit
IF EX(hold) EX MEM
I7 add x8,x8,x9
IF IF(hold) EX MEM x9 fwd
I8 addi x12,x12,-1
IF EX MEM
I9 bne x12,x0,loop
x12 fwd, branch taken in EX
IF EX MEM
I10 (wrong path)
IF (flush)
I1' lw x5,0(x10)
IF EX MEM
I2' lw x6,0(x11)
IF EX
I3' mul x7,x5,x6
IF
bubble
NOP
NOP NOP


clk
pc(S1)
0x00 0x04 0x08 0x0C (frozen) 0x10 0x14 0x18 (frozen) 0x1C 0x20 0x24 0x00 0x04 0x08
l1_state
IDLE HIT MISS WB FILL SWAP READ DONE IDLE MISS WB FILL SWAP READ DONE HIT IDLE HIT
Cache_stall
vc_hit
vc_swap
md_busy
stall_pc